Recap: Lecture 21

In Lecture 21, we

- Introduced and classified PDEs
- Introduced initial–boundary value problems
- "Derived" the 1D heat equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad \text{where} \quad \alpha = \kappa / c\,\rho > 0 \text{ "thermal diffusivity"}$$

Material	α [$\mathrm{cm}^2\,\mathrm{s}^{-1}$]
Gold	1.27
Iron	2.3×10^{-1}
Air	1.9×10^{-1}
Quartz	1.4×10^{-2}
Wood	8.2×10^{-4}

Lecture 22

In Lecture 22, we will:

1. Apply separation of variables to the 1D heat equation + initial + boundary conditions

Heat equation in 1D: Problem 1

Consider the temperature distribution in a rod of finite length L. The rod is initially heated to a prescribed temperature distribution $f(x)$, and then quenched in ice to maintain zero temperature at the ends. Assume the longitudinal sides of the rod are insulated, so that heat can only escape through the ends.

Governing PDE: 1D heat equation for temperature distribution $u(x,t)$ is

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \quad \text{for} \quad 0 < x < L$$

Initial condition:

$$u(x,0) = f(x) \quad \text{for} \quad 0 < x < L$$

Boundary conditions: Dirichlet BCs

$$u(0,t) = u(L,t) = 0 \quad \text{for} \quad t > 0$$

Separation of Variables (SoV)

This (and other) problems may be solved using the standard method of "Separation of Variables" (SOV).

Application of the SoV method to a 1D space + time PDE involves the following steps:

1. Separate variables to turn single PDE into multiple ODEs.
2. Apply boundary conditions to find eigenvalues/eigenfunctions.
3. Solve remaining ODEs for each eigenvalue.
4. Apply superposition principle to get a general solution in series form.
5. Apply initial conditions to find the series coefficients.

Solving the 1D heat equation using separation of variables

1. Separate variables to turn single PDE into multiple ODEs *which are tied together by a constant of separation* λ

We seek solutions of the separable form

$$u(x,t) = X(x)\,T(t)$$

$$\Rightarrow \quad X\,\dot{T} = \alpha\,X''\,T$$

$$\Rightarrow \quad \frac{\dot{T}}{\alpha\,T} = \frac{X''}{X} = \lambda$$

$$\Rightarrow \quad \dot{T} - \alpha\lambda\,T = 0 \quad \text{and} \quad X'' - \lambda\,X = 0,$$

which are two ODEs (one in time and one in space).

Solving the 1D heat equation using separation of variables

2. Apply the boundary conditions to find the eigenvalues and eigenfunctions. *This involves a systematic analysis of three possible cases for the constant of separation.*

The BCs give

$$u(0,t) = X(0)\,T(t) = 0 \quad \Rightarrow \quad X(0) = 0$$

and

$$u(L,t) = X(L)\,T(t) = 0 \quad \Rightarrow \quad X(L) = 0$$

We have the eigenvalue problem

$$X'' - \lambda\, X = 0, \quad X(0) = X(L) = 0.$$

For case $\lambda = k^2 > 0$:

$$X'' - k^2 X = 0 \quad \Rightarrow \quad X(x) = A\,\mathrm{e}^{kx} + B\,\mathrm{e}^{-kx}$$

and the BCs enforce $A = B = 0$, i.e., the trivial solution only.

For case $\lambda = 0$:

$$X'' = 0 \quad \Rightarrow \quad X(x) = A\,x + B$$

and the BCs enforce $A = B = 0$, i.e., the trivial solution only.

For case $\lambda = -k^2 < 0$:

$$X'' + k^2 X = 0 \quad \Rightarrow \quad X(x) = A\,\cos(kx) + B\,\sin(kx).$$

The BC $X(0) = 0$ gives $A = 0$, and then $X(L) = 0$ implies $k = k_n = n\,\pi\,/\,L$ $(n = 1, 2, \ldots)$ for non-trivial solutions. This gives the eigenfunctions and eigenvalues

$$X_n(x) = \sin(k_n x), \quad \lambda_n = -k_n^2 \quad (n = 1, 2, \ldots)$$

Solving the 1D heat equation using separation of variables

3. Solve the remaining ODEs for each eigenvalue.

We solve the ODE for T for the eigenvalues, such that

$$\dot{T}_n - \alpha\,\lambda_n\,T_n = 0 \quad \Rightarrow \quad T_n(t) = \mathrm{e}^{-\alpha\,k_n^2\,t}$$

for $n = 1, 2, \ldots$

This results in an infinite set of solutions of the form

$$u_n(x,t) = X_n(x)T_n(t) = e^{-\alpha k_n^2 t}\sin(k_n x) \quad \text{where} \quad n = 1, 2, 3\ldots$$

Solving the 1D heat equation using separation of variables

4. Apply Superposition Principle to get a general solution in series form.
   The series will have unknown coefficients to be determined.

$$u(x,t) = \sum_{n=1}^{\infty} C_n \, \mathrm{e}^{-\alpha k_n^2 t} \, \sin(k_n x)$$

5. Apply the initial conditions to find the series coefficients.

The IC gives

$$u(x,0) = \sum_{n=1}^{\infty} C_n \sin(k_n x) = f(x)$$

which is a Fourier since series for $f(x)$.

Hence

$$C_n = \frac{2}{L} \int_0^L f(x) \sin(k_n x) \, \mathrm{d}x$$

Solving the 1D heat equation using separation of variables

Let

$$f(x) = \begin{cases} x & 0 \leqslant x < \frac{L}{2} \\ L - x & \frac{L}{2} \leqslant x \leqslant L \end{cases}$$

We evaluated the coefficient in Ex 4.9 to be

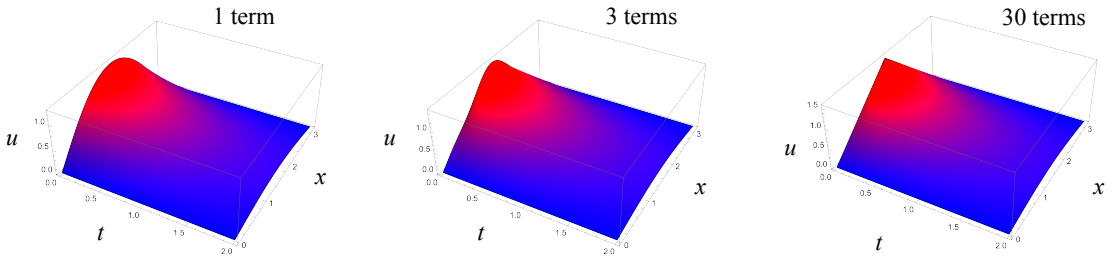
$$C_n = \frac{4\,L}{n^2\pi^2} \sin(n\pi/2)$$

Solving the 1D heat equation using separation of variables

The heat distribution along the rod is

$$u(x,t) = -\frac{4L}{\pi^2} \sum_{m=1}^{\infty} \frac{(-1)^m}{(2m-1)^2} \mathrm{e}^{-\alpha k_{2m-1}^2 t} \sin(k_{2m-1} x)$$

$$\alpha = 1, \quad L = \pi$$



The quartz case ($\alpha = 1\,\mathrm{mm}^2\,\mathrm{s}^{-1}$) and $L = \pi\,\mathrm{mm}$.

Solving the 1D heat equation using separation of variables

